

WILLOW CREEK CITY HALL

Feasibility Study — Installation of Solar Photovoltaic Panels on Municipal Buildings: City Hall, Willow Creek Elementary School, Municipal Gymnasium

Publication date: September 5, 2025 Sponsoring department: Office of the City Manager Prepared by: Municipal Public Works (Buildings and Energy Unit) with support from an external engineering firm, Sunwise Energy Studies

Executive Summary

This study assesses the technical, economic, and environmental feasibility of installing solar photovoltaic panels on the roofs of three key municipal buildings: City Hall, Willow Creek Elementary School, and the Municipal Gymnasium. The technical analysis confirms meaningful potential across all three sites, with an estimated usable roof area of approximately 10,225 sq ft (950 m²). Annual production could cover a significant share, estimated at 25% to 40%, of the buildings' current electricity consumption, generating substantial energy bill savings and reducing the municipality's carbon footprint. The initial investment is significant but could potentially be recovered over 12 to 16 years depending on grants and the selected self-consumption model. The study recommends continuing the project through detailed technical studies and active pursuit of external funding.

1. Introduction

In a context of energy transition and the need to control operating costs, Willow Creek City Hall wants to explore the potential for renewable energy production on its own building portfolio. This feasibility study aims to evaluate:

- The technical feasibility of installing photovoltaic panels on the roofs of City Hall, the elementary school, and the gymnasium.
- The potential energy production and expected self-consumption rate.
- The economic viability of the project, including costs, savings, and return on investment.
- The associated environmental benefits, including reduced CO2 emissions.
- Administrative and regulatory constraints.

2. Methodology

The study was conducted between May and August 2025 and was based on:

- Technical visits to the three buildings and visual roof inspections.
- Analysis of building plans, parcel data, and solar exposure estimates using simulation software.
- Collection and analysis of smart meter electricity consumption data for the three sites over the previous 24 months.
- Research on current photovoltaic technologies and costs.
- Preliminary consultations with installers and the local electric utility.
- Economic modeling based on different scenarios, including full or partial self-consumption with sale of surplus power.

3. Technical Analysis by Building

Building	Usable Roof Area (estimated)	Orientation / Pitch	Roof / Structure Condition	Installable Peak Capacity (estimated)	Comments
City Hall	1,615 sq ft (150 m²)	South / 30°	Good condition / likely OK	25 kWp	Minor shading from antennas.
Willow Creek Elementary School	4,305 sq ft (400 m², multiple roof planes)	Southeast, Southwest / 25°	Good condition / structure to verify	65 kWp	Requires detailed structural study of the roof framing.
Municipal Gymnasium	4,305 sq ft (400 m²)	Low slope metal roof	Average condition / structure to verify	60 kWp	Requires structural study and waterproofing review.
TOTAL	10,225 sq ft (950 m²)			150 kWp	

4. Estimated Energy Production Potential

- Estimated total annual production: approximately 165,000 kWh/year.
- Combined annual electricity consumption of the three buildings: approximately 500,000 kWh/year.
- Potential self-consumption coverage rate: 33%. This rate could be optimized through load management strategies, such as daytime electric vehicle charging.

5. Economic Analysis

- Estimated total investment cost: approximately \$180,000 to \$220,000 before tax, including equipment, installation, grid connection, and studies, excluding major structural reinforcement if required.
- City Hall: approximately \$30,000 to \$35,000.
- Elementary School: approximately \$75,000 to \$90,000.
- Gymnasium: approximately \$75,000 to \$95,000, including possible partial waterproofing work.
- Potential grants and incentives: to be investigated, including federal, state, county, and utility incentives, net metering or surplus sale mechanisms, and renewable energy programs. Grants are a key factor for improving project profitability.
- Recommended economic model: individual or shared self-consumption, if technically possible between nearby sites, with sale of surplus power.
- Estimated annual savings: \$25,000 to \$30,000 per year based on current tariff assumptions of roughly \$0.20/kWh, depending on actual self-consumption and surplus sale prices.
- Gross payback period without grants: estimated at 14 to 18 years.
- Net payback period with a 30% grant assumption: estimated at 10 to 13 years.
- Maintenance and insurance costs: estimated at approximately 1% of the initial investment per year.

6. Environmental Benefits

- Estimated annual CO2 emission reduction: approximately 15 metric tons of CO2 equivalent per year.
- Visible contribution to local and national energy transition goals.
- Strengthening of Willow Creek's environmentally responsible public image.

7. Regulatory and Administrative Aspects

- Building or planning applications may be required for each site depending on the final design and local permitting rules.
- Grid interconnection requests must be submitted to the local electric utility.
- Confirm whether City Hall or the school is located in a protected historic or design review district, in which case additional review may be required.
- Power purchase, net metering, or surplus sale agreements must be established as applicable.

8. Conclusions and Recommendations

Installing photovoltaic panels on the roofs of City Hall, Willow Creek Elementary School, and the Municipal Gymnasium appears technically feasible and offers meaningful energy and economic potential, although the project depends on additional structural studies, especially for the school and gymnasium, and on the availability of grants. The environmental benefits are significant.

It is recommended to:

1. Have the City Council formally validate the decision to continue the project.
2. Commission detailed technical studies, including structural diagnostics of the relevant roofs and precise panel layout design.
3. Actively monitor and pursue grant opportunities.
4. Refine the economic model, including individual or shared self-consumption and surplus sale options, based on study results and regulatory opportunities.
5. Include the necessary credits in the 2026 forecast budget for studies and, potentially, the beginning of work on a pilot building such as City Hall if the structure is confirmed suitable.

Issued in Willow Creek on September 5, 2025

Public Works Manager, [Manager Name]

Mayor, Mrs. Sparkle Merriweather